A unified processing account of historical German nominal and pronominal object order

This information theoretic corpus study novelly posits that processing facilitation was the core driver of word order variation in both *pronominal* double object constructions (i.e., those with two pronominal objects), (1), and *nominal* ones (i.e. two nominal objects), (2), in historical German.

Introduction. In historic German and modern Standard German, nominal double object constructions differ in whether the morphologically dative-case marked indirect object precedes the morphologically accusative-marked direct object, (1a), or vice versa, (1b). This word order variation has been much studied and proposed to be determined by a range of processing-related factors (givenness, animacy, information uniformity, unmarked orders; e.g. [1]). In contrast, pronominal double object constructions in modern Standard German display a near-rigid accusative-before-dative order, (2b) (cf. (2a)). Pronominal double object constructions in historic German did, however, also display word order variation. Yet, the single previous account of this variation ([2]) does not relate it to processing. The current study thus aimed to determine whether a uniform account of the two double object construction types was possible.

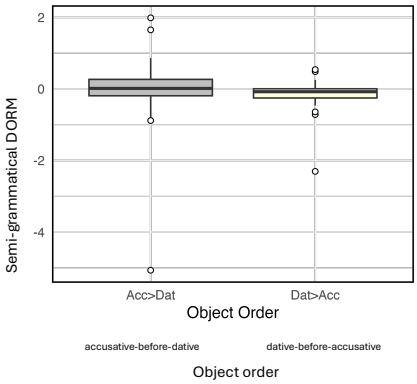
Method. All pronominal double object constructions were extracted from the 600,000-word GerManC corpus (1650-1800; [3]; n=61, dative-before-accusative n=33, 54%). For each relevant sentence, information uniformity was calculated since, according to the Uniform Information Density hypothesis ([4]), maximising uniformity in surprisal – the probability of a linguistic unit appearing in a certain context ([5], [6]) – across utterances (i.e. information uniformity) facilitates processing. Here, information uniformity was determined with the *Deviation of the Rolling Mean* (DORM) statistic ([7]), calculating the variance in the means of the unigram surprisal values of all adjacent pairs of items. Further, both a uniquely lemma-based, lexical measure and a novel semi-grammatical measure of DORM were used, the latter based on the word forms of pronouns and lemmas of other forms. This contrasts with previous historical corpus studies, which used lexical DORM only, whilst still addressing the data sparsity and orthographic variation in historic data (e.g. [1], [8]). To ascertain the influence of object order, lexical and semi-grammatical DORM values for the artificially created, alternative non-attested order were then subtracted from the respective values for the original attested order for each sentence, following [9]. Further, it was expected that any processing advantage should relate to full verb position, following [11]/[1], and the referent's accessibility, namely the ease of its retrieval from memory (e.g. [10]). Accessibility can be influenced by pronoun animacy and salience difference, namely the relative distance of the objects from their previous mentions (antecedents). Additionally, cliticisation was examined as it was previously identified as relevant for pronominal double object constructions ([2]).

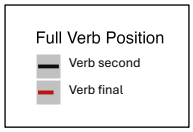
Results. No lexical DORM difference was found between the two orders but, crucially, a semi-grammatical DORM difference was: sentences which attested the dative-before-accusative order were significantly more uniform in that order than in the alternative, accusative-before-dative order, (3) (Wilcoxon signed rank test: $V = 147$, $p < 0.05$). Further, according to Firth's logistic regression (R-package *logistf*; [12]), (4), accusative pronoun type (clitic vs. non-clitic) is irrelevant (cf. [2]) but there is crucially a tentative influence of the dative pronoun type ($1^{st}/2^{nd}$ person vs. 3rd. person) and an influence of salience difference (less distant dative antecedent) on the likelihood of the dative-before-accusative order. This suggests a grammatical (cf. lexical) processing advantage of that order, explainable by greater accessibility of the dative referent. Further, there was also an interaction between lexical DORM and full verb position: likelihood of the dative-before-accusative order increases in lexically less uniform cases when the full verb is in final (cf. second) position, (5). Following [1], this follows as verb valency information comes late with a final position verb, but early presentation of a dative object (cf. an accusative one) disambiguates verbs as ditransitive, easing processing. Finally, instances of the accusative-before-dative order can be explained by processing instead being facilitated by an unmarked order-based strategy (e.g. [13]).

(1) a. Robin gab **dem Mann das Buch**. (Modern German, dative-before-accusative)
Robin gave the man the book
b. Robin gab **das Buch dem Mann**. (Modern German, accusative-before-dative)
Robin gave the book the man
'Robin gave the man the book.'

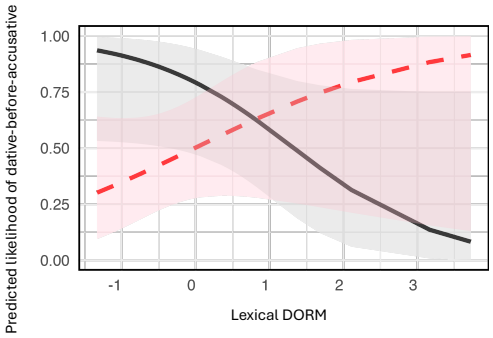
(2) a. **??**Robin gab **ihm es** . (Modern German, dative-before-accusative)
Robin gave him it
b. Robin gab **es ihm** . (Modern German, accusative-before-dative)
Robin gave it him
'Robin gave him it.'

(3)





(5)



(4) Logistic regression results

	Coef.	SE (Coef.)	Lower 0.95	Upper 0.95	Chisq	p
Intercept	0.10	0.64	-1.75	1.50	0.019	0.89
Lexical DORM	-0.18	0.38	-1.15	0.92	0.15	0.70
Full verb position	1.33	0.89	-3.89	0.49	2.04	0.15
Dative pronoun	-2.07	1.21	-6.07	0.26	3.00	0.08 .
Accusative pronoun	-0.68	0.80	-2.47	0.92	0.69	0.41
Salience difference	-0.12	0.06	-0.31	-0.01	5.23	0.02 *
Leixcal DORM × Full verb position	1.64	0.78	0.26	4.08	5.72	0.02 *

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